Stat 5200: Analysis of Designed Experiments

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Introduction

Objectives and overview

- This material is based Chapters 13–14 from the textbook "A First Course in Design and Analysis of Experiments" (First Edition) (Oehlert, 2000).
- Chapter outcomes include:
  - You will be able to identify and design a block design.
  - You will be able to describe strengths and weaknesses of block-designs.
  - You will be able to create and analyze the most common types of block designs, including completely randomized blocks and Latin Squares.

Introduction: beyond the CRD

- Most recently, we have focused on the analysis of data using ANOVA models.
- We will now discuss specific experimental designs.
- Fortunately, the skills built for analyzing data will remain relevant.
- Most design principles build on the completely randomized design, but add additional structure to the randomization procedure.
- The additional structure is often practical and natural, but can also provide an analytic advantage by reducing variance among observations.

Introduction: beyond the CRD II

- Recall how a completely randomized design (CRD) is specified:

Randomized Complete Block Designs

Blocking

- While considered the "gold standard", CRDs are not always possible or practical.
- In this module, we introduce the idea of blocking.

Blocking: basic idea

- Choose a set of experimental units that are similar in some way to form a "block".
- Because things within a block are similar, the variance within-blocks is reduced.
- Blocks are often natural distinctions: units from a specific supplier, units spatially close in a field, or units taken during similar times.

The Randomized Complete Block Design

- There are many types of block designs. We will focus first on complete block designs.

Randomized Complete Block Designs (RCB)

- There are g treatments.
- There are r blocks.
- Each block receives exactly 1 of the g treatments, for a total of $N = rg$ experimental units (EUs).
- Within each block, the treatment assignment for each EU is randomized.

CRD vs RCB

- How does a randomized complete block design differ from a completely randomized design?
  - CRD: All N EUs can be randomly assigned to any treatment.
  - RCB: We first pick / identify r blocks, which consist of g EUs each. Then, within each block, we randomly assign the g treatments.

CRD vs RCB II

Example: batch-based blocks

Suppose you work at a bakery, and you have r different farms that supply you with eggs. You are interested in testing new baking times for your recipe: $T1$, $T2$, $T3$, $T4$.

CRD vs RCB III

Example: temporal blocks

Suppose you work in a laboratory, and you would like to test three different chemical catalysts: $C1$, $C2$, $C3$. Your analysis instrument must be recalibrated each morning, and measurements are timely and expensive, so you take three measurements per day over r days.

CRD vs RCB IV

Example: spatial blocks

Suppose you work at an artisanal creamery and age cheese wheels in a climate-controlled cave with r different vertical shelves. Because temperature and humidity vary from the floor to the ceiling, you treat each shelf as a block. You are testing four different brine washes: $W1$, $W2$, $W3$, $W4$.

Complex example: Mealybugs (Ex 13.1)

- This next example shows a blocked design, and helps us remember the definition and importance of identifying experimental units. From Module 1:
- *Experimental Units* (EU) are the entities which a treatment will be applied.
- *Measurement Units* are the individual or piece of experiment material you actually measure.

Complex example: Mealybugs (Ex 13.1) II

Mealybugs on cycads

Zookeepers wish to test three different treatments to combat the spread of mealybugs: water (a control), horticultural oil (a standard non-toxic pesticide), and fungal spores.

- Five infested cycads (plants) are moved to a testing area. Three branches are randomly selected from each.
- Each branch is randomly assigned to one of the treatment groups.
- On each branch, two 3cm $\times$ 3cm patches are identified.
- Mealybugs are counted on both patches, treated, and then recounted after 3 days.

Complex example: Mealybugs (Ex 13.1) III

	Plant				
	1	2	3	4	5
Water	-9	18	10	9	-6
	-6	5	9	0	13
Spores	-4	29	4	-2	11
	7	10	-1	6	-1
Oil	4	29	14	14	7
	11	36	16	18	15

Figure 1: Table 13.1 from Oehlert (2000).

Complex example: Mealybugs (Ex 13.1) IV

- Question: What type of design is this? What are the experimental units, and measurement units?

Common counter example

- A common mistake that is made is believing a study is a block design, when it is not.
- Below is a such an example. It is NOT a block design, even though there are natural "blocks".

NOT a block design: pig pens

A farmer would like to test the effect of 3 different feeds on pig weight. The farmer has r pens of pigs, each with 2 pigs. The farmer randomly assigns each of the pens with one of the three different feeds. They record the change in weight of the individual pigs.

Common counter example II

- Question: What type of design is this? What are the experimental units, and measurement units?

Block design: summary

- Blocks of EUs are often natural (time, space, batch, etc). However, natural grouping does *not* imply a block design was used (e.g., assigning feed to pig-pens).
- Block designs are carefully planned, not haphazard. That is, for each block, we randomly assign a EU in the block exactly one of the g treatments. The number of blocks is unconstrained.
- Block designs refer to how randomness is performed. We can still have complete treatment structures, like factorial treatments, within each block.

Block design: summary II

- When and why do we block, versus CRD?
  - We block to increase power and precision. This is done by decreasing the error variance, which is achieved by finding groups of units that are homogeneous.
  - The RCB is an effective design where there is a single source of extraneous variation in the response that we can identify ahead of time and use to partition the units into blocks.
  - Blocking is done at the time of randomization; you can't construct blocks after the experiment has been run.

How to analyze RCB designs

- The ANOVA models we have used are also useful for RCB; we don't have to "re-invent the wheel".
- The exact same models, assumptions, software, and code will be used to analyze data from an RCB; we just have to interpret things a little differently.

Model for RCB designs

The standard model for the ith treatment in the jth block will be:

$$y_{ij} = \mu + \alpha_i + \beta_j + \epsilon_{ij}, \quad \epsilon_{ij} \overset{iid}{\sim} N(0, \sigma^2)$$

In this model,

- i: the index for treatment level. Ranges from $1, \ldots, g$.
- j: the index for the block. Ranges from $1, \ldots, r$.
- In this model, there is no interaction effect. This is both because there is not enough data to estimate the effect, but also because we believe the blocks themselves to have little impact on the treatment effect.
- This is very similar to a analyzing a two-way factorial analysis without replication.

Model for RCB designs II

- Our Sum-of-Squares breakdown is:

$$SS_{Total} = SS_{Treatment} + SS_{Block} + SS_{Error},$$

  That is, to see the treatment effect, we first account for the block-effect, which is how we reduce the variance among measurements.
- The primary difference between unreplicated 2-way ANOVA: We do not care about the block effect, and we cannot perform inference because the blocks are not random.

Mealybugs example

- Consider the Mealy-bugs example. Below are the data, where the response for each patch has been averaged.

Table 1: Average change in mealybug after each treatment for each EU (columns are plant).

	1	2	3	4	5
Water	-7.5	11.5	9.5	4.5	3.5
Spores	1.5	19.5	1.5	2.0	5.0
Oil	7.5	32.5	15.0	16.0	11.0

Mealybugs example II

```
plants <- read.csv('mealybugs.csv')
str(plants)

'data.frame': 15 obs. of  3 variables:
 $ trt       : chr  "Water" "Spores" "Oil" "Water" ...
 $ plant     : int  1 1 1 2 2 2 3 3 3 4 ...
 $ avgchange: num  -7.5 1.5 7.5 11.5 19.5 32.5 9.5 1...
```

Mealybug example (continued)

```
mod <- aov(avgchange ~ factor(plant) + trt, data = plants)
summary(mod)

                Df Sum Sq Mean Sq F value  Pr(>F)
factor(plant)  4  686.4  171.60   9.681 0.00371 **
trt            2  432.0  216.02  12.187 0.00373 **
Residuals      8  141.8   17.73
---
Signif. codes:
0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Mealybug example (interpretation)

- In the results above, R outputs a F-value and P-value for the block factor (`plant`). Again, measurements are not randomly assigned to blocks, so we will ignore this output.
- More importantly, we see a significant effect for treatment. That is, there is fairly strong evidence for differences in mealybug counts between the treatments.
- We should continue to check model assumptions; doing so does not indicate there are violations of our key assumptions.
- We may be interested in what the estimated treatment means are, after accounting for variance among the blocks. This is easily accomplished with the `emmeans` package and function.

Mealybug example (interpretation) II

```r
library(emmeans)
# Get treatment average, after adjusting for block effect.
em_mod <- emmeans(mod, ~ trt)
```

```
 trt     emmean   SE df lower.CL upper.CL
 Oil       16.4 1.88  8  12.0582    20.74
 Spores     5.9 1.88  8   1.5582    10.24
 Water      4.3 1.88  8  -0.0418     8.64

Results are averaged over the levels of: plant
Confidence level used: 0.95
```

Mealybugs: further analysis

- Just like any of our previous examples, we may want to perform tests for sets of contrasts.
- A natural set of contrasts in this case may be: all pairwise comparisons, in which case, Tukey's adjustment is a good choice to control for multiple comparisons.

Mealybugs: further analysis II

```
pairs(em_mod)
```

```
 contrast       estimate   SE df t.ratio p.value
 Oil - Spores       10.5 2.66  8   3.943  0.0106
 Oil - Water        12.1 2.66  8   4.544  0.0047
 Spores - Water      1.6 2.66  8   0.601  0.8236

Results are averaged over the levels of: plant
P value adjustment: tukey method for comparing a family of 3 est
```

Unbalanced blocks (missingness)

- We have assumed a complete block design, meaning each block sees each treatment exactly once.
- In practice, it can be common to have missing data.
- In this case, missingness causes issues for a block-analysis in the same way as an unbalanced factorial analysis: our SS terms are no longer independent.
- Just as before, we need to consider the difference between Type I (sequential SS), Type II (Hierarchical), and Type III (Fully Adjusted). Any type works, but think carefully about what you are doing.

Is blocking worth it?

- Aside from sometimes being a natural thing to do, there are two primary consequences for blocking in the analysis:
  - Good: Blocking can result in increased precision, by grouping observations into similar blocks. This reduces σ^2, and increases statistical power if done correctly.
  - Bad: If we blocked, we have to account / estimate the block effect. This reduces the degrees of freedom available for estimating the treatment effect! This has a negative effect on statistical power.
- Thus, there is a competing impact in blocking. If the blocks don't actually differ from each other, blocking *will* cost us degrees of freedom without reducing the sum of squares, so it can actually be worse!

Is blocking worth it? II

- Whether or not the blocking was useful can be estimated. See Chapter 13.2.3 of Oehlert (2000) for details on how to compute this. Maybe not useful, since you need the results of an experiment to estimate how useful blocking in the experiment would be.
- **Summary:** only block if there is reason to believe that there are homogenous groups of EUs. **We cannot block after obtaining the data!**

Latin Squares and Crossover Design

Additional Topics

References and Acknowledgements

Oehlert GW (2000). *A First Course in Design and Analysis of Experiments*. W. H. Freeman. ISBN 0-7167-3510-5.

- Compiled on October 16, 2026 using R version 4.5.2.
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- We acknowledge students and instructors for previous versions of this course / slides.